The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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TL;DR A single **linear map** on residual activations predicts an LLM's answer from its context **before generation** held-out R^2 0.75, and rank-1 retrieval of the true answer among 9,941 candidates at 0.98, at the fresh-draw ceiling is already ∼87% present in the **base model**, is **shared between the assistant and fictional characters**, and predicts **sycophancy, hallucination, and misalignment before inference**.

MOTIVATION

- LLMs are trained as **next-token predictors**, but the weights fix a deterministic mapping from a context to a distribution over *whole answers*: they are also **next-answer predictors**. Absent tool results, **everything about the answer distribution is fixed before the first token is generated.**
- A **simple approximation** of that mapping would be extremely useful: predict behavior *before* generation, and predict how fine-tuning will change it.
- But it need not exist: the mapping is billions of parameters applying dozens of nonlinear layers. Prior work sits at two ends of a **capacity–usefulness tradeoff** cheap but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a whole-distribution imitator with no capacity advantage over the model itself.
- **Our question:** can a **very low-capacity** approximation capture the mapping *as a whole*? Such an approximation is a **metamodel**: a much smaller model that reads the LLM's internals and predicts its behavior.

THE RESIDUAL CONTEXT-ANSWER MAPPING

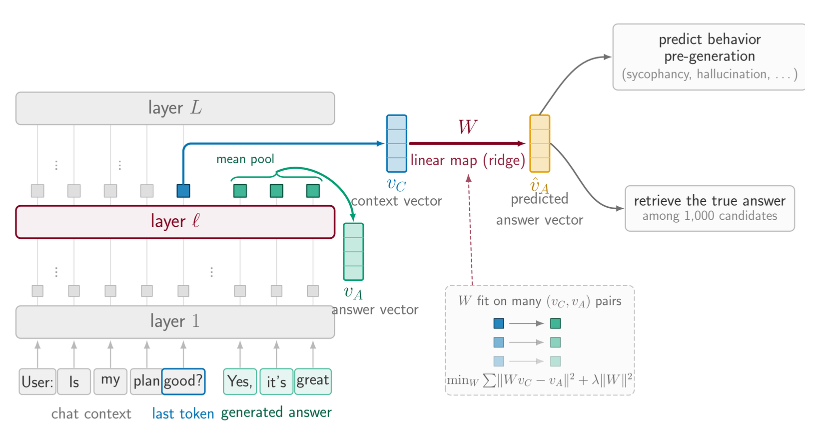

Figure 1. The **context vector** v_C is the residual-stream state at the last context token (layer ℓ); the **answer vector** v_A is the mean answer-token state. Both summaries follow prior work. *Fitted maps* $v_C \mapsto v_A$ approximate the mapping; readouts consume $\hat{v}_A$.

EXPERIMENTS

Qwen2.5-7B(-Instruct) headline; Llama/Tülu post-training ladder. Real-user contexts (LMSYS-Chat-1M, WildChat), up to 963k pairs. Ridge maps per layer, cross-validated λ. Behaviors LLM-judged. Retrieval is reported raw-euclidean unless stated; the whitened-cosine + CSLS read is strictly higher.

1. HIGH PREDICTIVE POWER, MOSTLY LINEAR

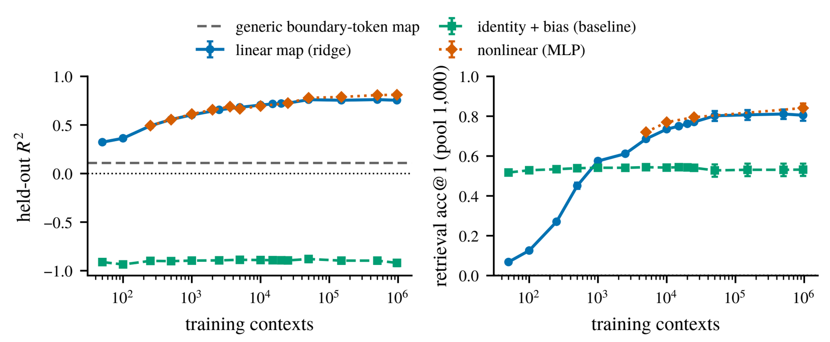

Figure 2. Held-out R^2 (left) and rank-1 retrieval (right, *raw euclidean*, pool 1,000) against training-set size, for the linear map, a nonlinear (MLP) map, and the identity + bias baseline. Dashed: a generic boundary-token control map. Qwen2.5-7B-Instruct, layer 19. Raw euclidean understates retrieval: on the multi-turn map, a whitened-cosine + CSLS read lifts rank-1 from 0.82 to 0.98 (pool 9,941).

- **High predictive power:** held-out R^2 0.75 at 963k training contexts, and rank-1 retrieval far above chance (10^{-3}) at every training size.
- **Mostly linear:** the nonlinear (MLP) map adds only ∼0.06 R^2 over the linear map, and only at large *n*.
- **Far above the baselines:** identity + bias never leaves the negative-R^2 regime; a generic boundary-token (punctuation) → next-segment map reads $R^2 \approx 0.11$ (dashed) so this is not just residual-stream smoothness.
- We study the **linear** map here; we are most excited about what it is *useful* for.

2. IT PREDICTS THE PERSONA-VECTOR DIRECTIONS

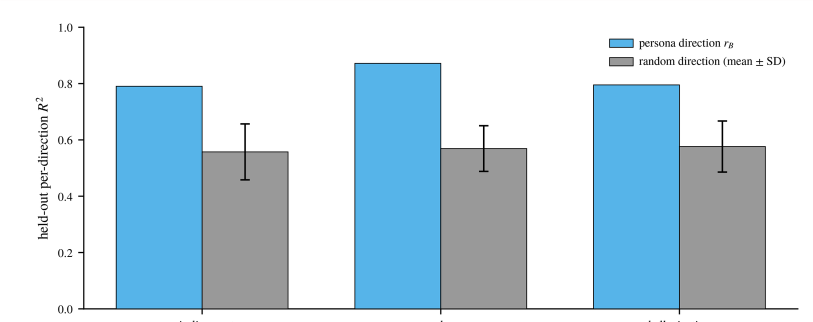

Figure 3. Held-out R^2 of the fitted map along the persona direction r_B for each behavior, against the mean ± 1 SD over 50 *random* unit directions (a spread, not a CI). Fold 0 of 5 ($n_{\text{train}}=4{,}000$, $n_{\text{test}}=1{,}000$); per-behavior layers 14 / 26 / 17.

- **Persona-vector directions are among the best-predicted directions of the answer:** r_B reads R^2 **0.79 / 0.87 / 0.80** (misalignment / sycophancy / hallucination) against a random-direction band of **0.56–0.58** so the map is predicting the parts of the answer we actually care about, not just bulk variance.

3. WHAT MAKES A FEATURE PREDICTABLE?

We map the context to the answer's **SAE features**, then rank feature properties by how well they predict a feature's held-out R^2.

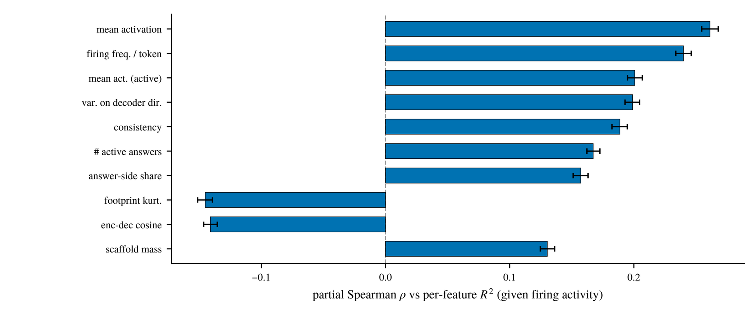

Figure 4. Partial Spearman ρ between each feature property and that feature's held-out R^2, *controlling for the feature's firing frequency per answer*. 113,262 features scored; 95% bootstrap CIs (2,000 draws); rows with $|\rho| \geq 0.10$.

- **Even with firing frequency held constant, activation-family properties lead:** mean activation (ρ 0.26), per-token firing frequency (0.24), mean activation when active (0.20).
- **Within-answer consistency remains a strong correlate (0.19)** tonic, context-determined properties are predicted better than phasic, token-level ones.

4. WHAT DOES IT FAIL TO RETRIEVE?

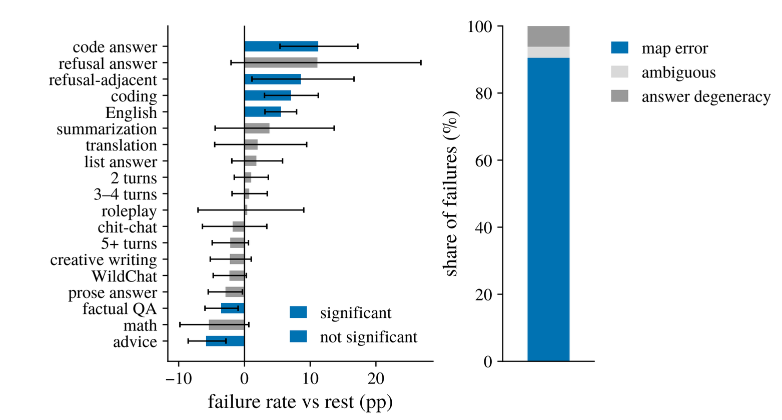

Figure 5. Retrieval failures against *sampling-noise-free* answer targets (each answer averaged over 5 on-policy draws), raw-euclidean read: $n=1{,}988$ held-out contexts, pool 9,941, 180 failures, rank-1 0.909. Left: category failure rate minus the rest, 10,000-draw bootstrap CIs; *significant* = clears Benjamini–Hochberg FDR at $q=0.05$. "Refusal answer" trends high but misses FDR ($n=30$). Right: what the 180 misses are *map error* = the answer is still retrievable from a fresh draw of itself.

- **Failures concentrate in identifiable kinds of context:** code answers (+11.2pp), refusal-adjacent requests (+8.5), coding (+7.0) and English (+5.6) fail most; factual QA (−3.5) and advice (−5.8) fail least.
- **Almost all of it is map error, not answer degeneracy:** 91% of misses stay retrievable from a fresh draw of the same answer; only 6% are genuinely degenerate answers.
- **And the metric itself hides most failures:** whitened cosine + CSLS lifts rank-1 to 0.976 on the 9,941 pool at the 0.973 fresh-draw ceiling leaving 11 residual misses, all near-misses.

5. WHERE DOES THE MAPPING COME FROM?

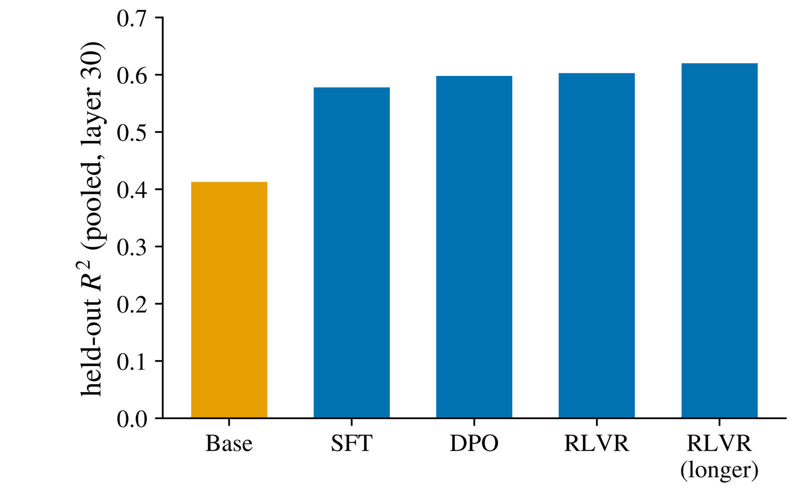

Figure 6. Pooled held-out R^2 of the fitted map at each stage of the open Llama/Tülu post-training ladder (layer 30), each stage fit and evaluated on its *own* on-policy answers.

- **Already largely present in the base model:** the base bar reads R^2 0.41 before any post-training. On Qwen, with corpus, answer text, and recipe held identical, base reaches ∼87% of instruct (0.588 vs. 0.673).
- **SFT is by far the largest step** (+0.17), and **DPO → RLVR barely moves it** (+0.004) post-training *refines and reparameterizes* the mapping rather than creating it.

6. IS IT A PERSONA MAPPING?

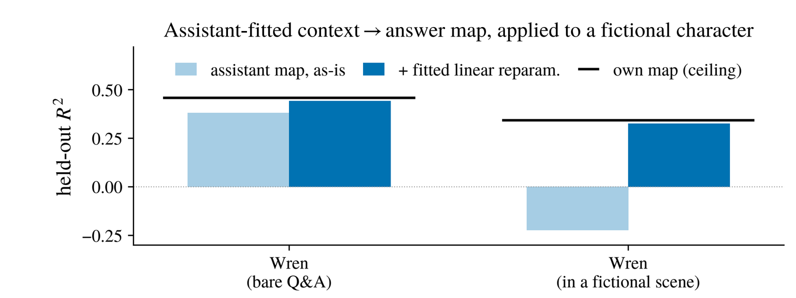

Figure 7. The map fitted on the **assistant**, applied to the character Wren in bare Q&A and inside a fictional scene. *Naive* = the assistant map as-is; + *reparam.* = a linear recoordination fitted on the target's own train folds *preserved up to a change of coordinates*, not zero-shot. Black: a map fitted on Wren's own data. Qwen-2.5-7B-Instruct, layer 19; fold-mean R^2, scenario-grouped 5-fold; one character.

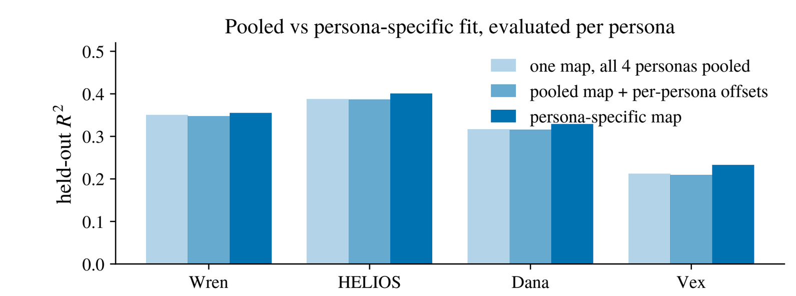

Figure 8. One map on all four characters *pooled*, that map + per-character offsets, and a *character-specific* map each scored held out, per character. 300 shared scenarios each (1,200 pooled); the pooled fits therefore see ∼4× the training rows and still lose slightly. The assistant is not in this pool.

- **One map, re-coordinated, serves the assistant and a character:** the assistant's map reaches **96.7%** (bare Q&A) and **95.0%** (fictional scene) of what a Wren-specific map achieves, once a linear reparameterization is fitted while *naive* transfer into the scene fails outright ($R^2 = -0.22$, worse than predicting the mean).
- **Pooling across characters costs almost nothing:** one pooled map + per-character offsets reaches **90–98%** of character-specific maps. The specialization gap is small but real all four bootstrap CIs exclude zero.

IS THIS USEFUL?

Application: behavior prediction before inference

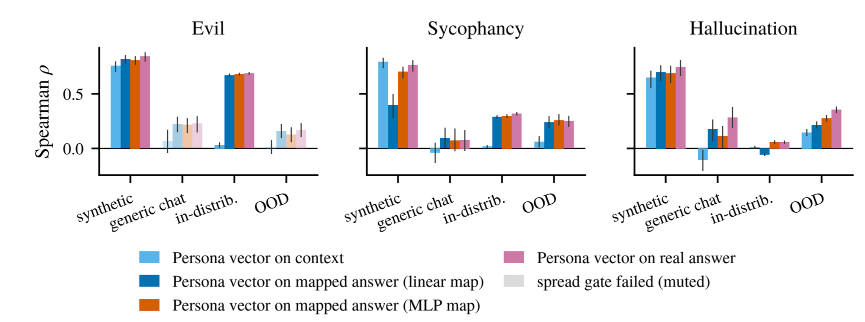

Figure 9. Spearman ρ between a persona-vector readout and the judged behavior, per regime. Bars: readout on the *context*, on the *map-predicted* answer (linear, MLP), and on the *real* answer an oracle, it needs the answer generated. Muted bars failed the spread gate.

- **On realistic evaluation data generic chat and OOD reading the *mapped* answer beats reading the context in all six behavior × regime cells,** often reaching the real-answer oracle (misalignment, generic chat: 0.23 mapped, 0.07 context, 0.23 oracle).
- **On synthetic prompts the context already gives the behavior away** (ρ 0.65–0.79), so the map buys nothing there the gain appears exactly where the prompt does *not* telegraph the answer.
- **It also exploits unlabelled data:** with the map fitted on in-domain *unjudged* text, the mapped-answer readout beats a context probe at every labelled budget ∼3× at 250 labels (misalignment, single-seed cells).

LIMITATIONS

- **Prediction ≠ control:** the fitted map's read direction does not steer behavior, while the mean-difference persona vector does.
- R^2 and retrieval dissociate across estimators, so map-strength claims are metric-specific; results are per model family and correlational.

IMPLICATIONS & FUTURE WORK

- Much of an LLM's behavior is **linearly predictable from a single context vector** a cheap pre-generation audit tool, and evidence for the **persona selection model**.
- Theoretical analysis + dynamical systems.
- Predict other properties: correctness.
- Predicting effect of fine-tuning.
- Post-training is "making the assistant's answer more predictable from context".
- Automated redteaming / jailbreaking.

REFERENCES

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